

## Lab 2 Report

DS5030

Jisung Lee, Jacob Ellis

### Recommendation

We are using cosine similarity with binary vectors. Instead of counting the total number of words used in each song, we record whether or not the song uses each of the words. This scored a precision@3 of 0.236 on the Top-40 benchmark, higher than the three baseline methods tested in the lab previously.

### Evidence

The test takes 560 songs, scores each against the full library, drops the song itself and everything by the same artist, keeps the top three matches, and records how many share the query's genre.

| Method        | Precision@3 |
|---------------|-------------|
| Dot           | 0.092857    |
| Cosine        | 0.191071    |
| tfidf_cosine  | 0.210119    |
| Cosine binary | 0.346310    |

The largest effect is caused by normalization; moving from dot product to cosine roughly doubles the score. TF-IDF adds a slight improvement, taking cosine similarity from 0.191 to 0.210, and the binary flag takes it to 0.236, which is a 12% relative improvement over the best of the three baselines.

### Song Evidence

The query song throughout the process was Gary Moore's "What Are We Here For," a blues track with 218 words, with "we" (27 times), "are" (10), and "for" (9) being the most frequent words. The Prodigy's "We Gonna Rock" was the top dot-product match to the query. Its total

score against the query was 2,136, and 2,106 of that came from the word "we." One grammatical word caused most of the similarity on its own, which shows a lack of real similarity. Lauryn Hill's "Interlude 5" was second for the dot product. At 1,903 words, it is nearly nine times larger than the query. A track with a significant number of words has enough function words to score high against almost any song in the library, so this reflects size more than its actual content. In the binary test, both of these songs fall out of the top of the rankings entirely.

### **Limiations**

Lyric word counts ignore the actual sound of a song. Two songs from very different genres can have the same everyday vocabulary and share no actual similarity. We see that when the query is a blues song, the cosine matches are metal and punk, signaling a weak connection. Adding other features like tempo, key, loudness, and energy would consider other aspects of a song that may signal similarity better.